Math 142, F20 **Name:** Answer

Quiz 5

<u>Instructions:</u> **There are two problems. You must show all of your work to receive credits for correct answers.**

Problem 1. (6 points) Determine whether the following improper integrals are convergent or divergent and evaluate those that are convergent.

(a) $\displaystyle\int_1^{+\infty} e^{-x/3}\,dx$

$$= \lim_{b\to+\infty}\int_1^b e^{-x/3}\,dx$$

$$= \lim_{b\to+\infty}\left[-3e^{-x/3}\right]_{x=1}^{x=b}$$

$$= \lim_{b\to+\infty}\left[-3e^{-b/3}+3e^{-1/3}\right]$$

$$=(-3)\cdot 0+3e^{-1/3}$$

$$=3e^{-1/3} \Rightarrow \text{converges!}$$

(b) $\displaystyle\int_{-\infty}^{\pi} \cos(2\theta)\,d\theta$

$$= \lim_{a\to-\infty}\int_a^\pi \cos(2\theta)\,d\theta$$

$$= \lim_{a\to-\infty}\left[\frac{1}{2}\sin(2\theta)\right]_{\theta=a}^{\theta=\pi}$$

$$= \lim_{a\to-\infty}\frac{1}{2}\left[\sin(2\pi)-\sin(2a)\right]$$

$$= \lim_{a\to-\infty} -\frac{1}{2}\sin 2a \to \text{DNE}$$

$$\Rightarrow \text{diverges!}$$

(c) $\displaystyle\int_1^3 \frac{1}{\sqrt{x-1}}\,dx$

$\frac{1}{\sqrt{x-1}}$ is not defined at $x=1$.

Thus $\displaystyle\int_1^3 \frac{1}{\sqrt{x-1}}\,dx = \lim_{c\to1^+}\int_c^3 \frac{1}{\sqrt{x-1}}\,dx$

$$= \lim_{c\to1^+}\left[2\sqrt{x-1}\right]_{x=c}^{x=3}$$

$$= \lim_{c\to1^+} 2\left[\sqrt{2}-\sqrt{c-1}\right]$$

$$= 2\sqrt{2} \Rightarrow \text{converges!}$$

Problem 2. (4 points) Use the trapezoidal rule and Simpson's rule respectively with $n = 4$ to eatimate the integral $\int_0^2 x^2 + 1\, dx$.

$a=0,\ b=2,\ n=4,\ \Delta x = \frac{b-a}{n} = \frac{2-0}{4} = \frac{1}{2}$

i	0	1	2	3	4
x_i	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2
$y_i = f(x_i)$	1	$\frac{5}{4}$	2	$\frac{13}{4}$	5

$$T = \frac{\Delta x}{2}(y_0 + 2y_1 + 2y_2 + 2y_3 + y_4)$$

$$= \frac{(\frac{1}{2})}{2}\left(1 + 2\cdot\left(\frac{5}{4}\right) + 2\cdot(2) + 2\cdot\left(\frac{13}{4}\right) + 5\right)$$

$$= \frac{1}{4}\cdot 19 = \frac{19}{4}$$

$$S = \frac{\Delta x}{3}(y_0 + 4y_1 + 2y_2 + 4y_3 + y_4)$$

$$= \frac{(\frac{1}{2})}{3}\left(1 + 4\cdot\left(\frac{5}{4}\right) + 2\cdot(2) + 4\cdot\left(\frac{13}{4}\right) + 5\right)$$

$$= \frac{1}{6}\cdot 28 = \frac{14}{3}$$